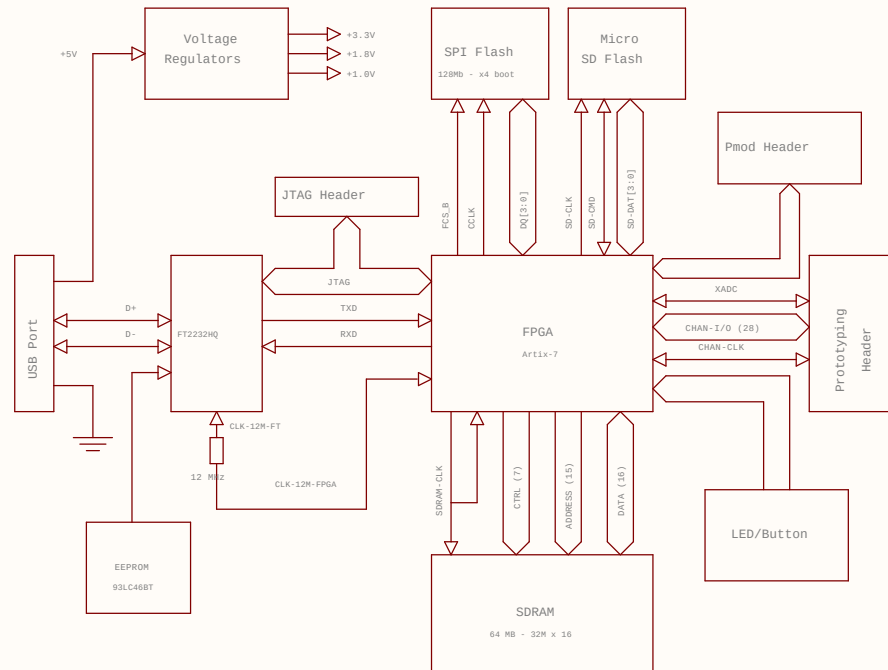
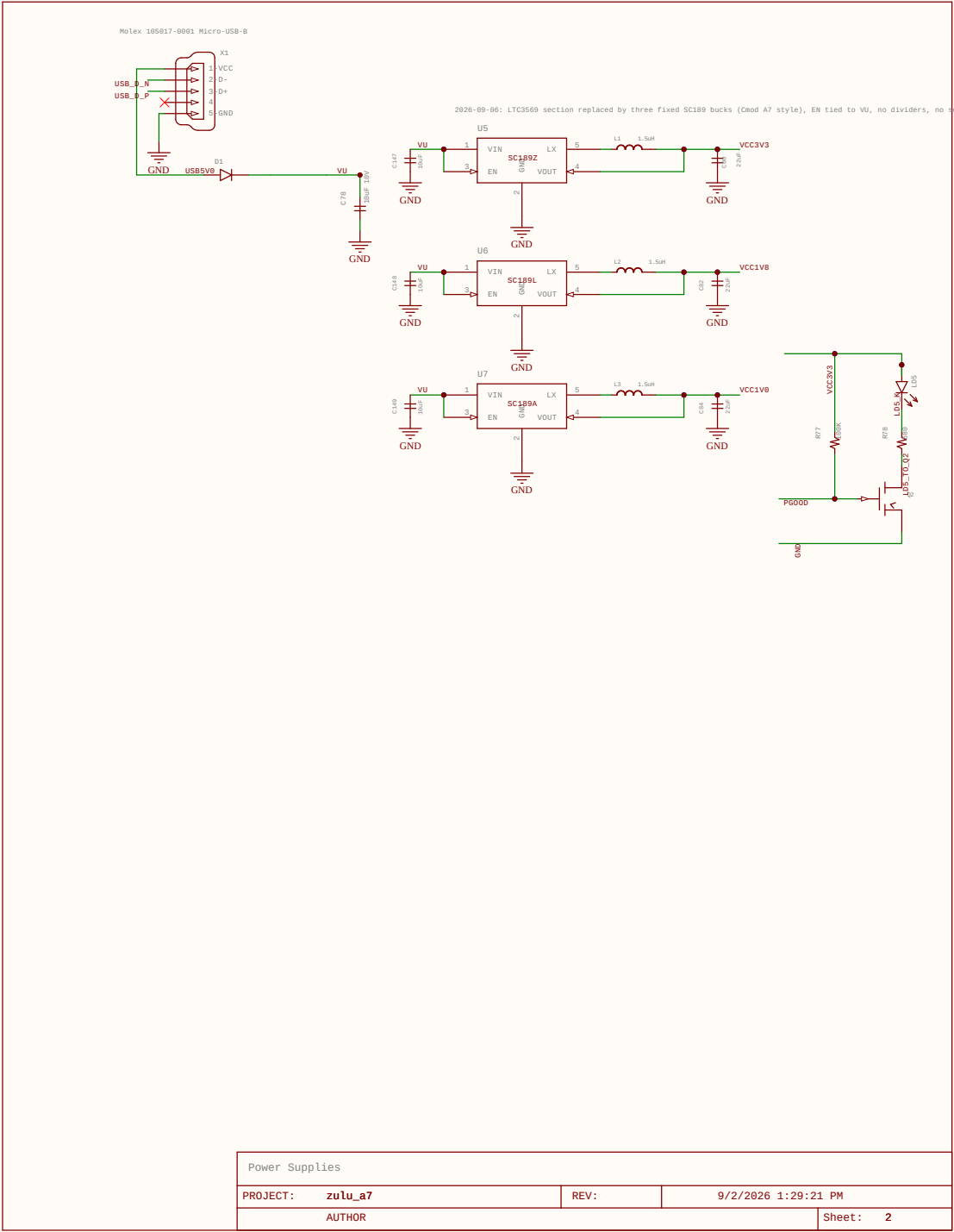


SPI flash sits on the FPGA's dedicated master-SPI config pins: CCLK C11, FCS_B K19, DQ[3:0] on D18/D19/G18/F18. Read at power-up.

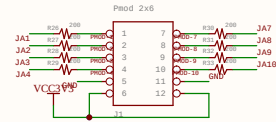
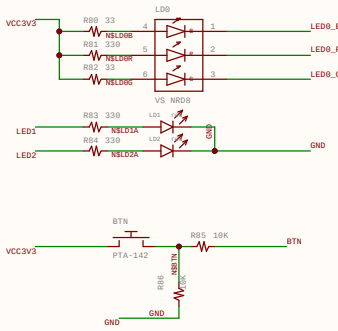
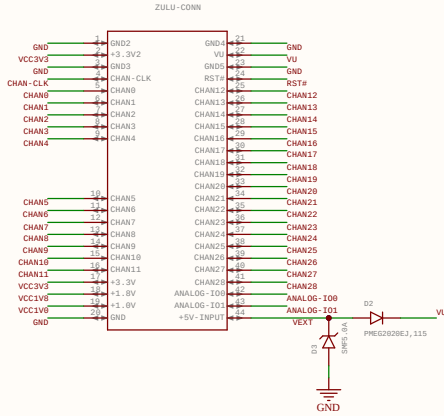
microSD runs the native 4-bit SD bus, not SPI: CLK, CMD and DAT[3:0] on six bank-34 balls of it's own, so a card cannot touch the config bus.



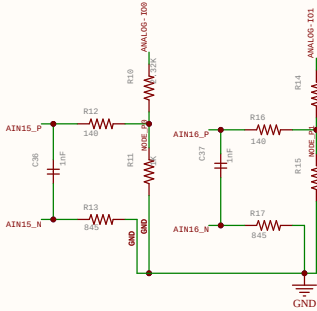
Block Diagram			
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X2: 44 pins on a 2.54 mm grid, rows 0.700 in apart, board 69.85 x 20.32 mm.
Numbered right to left and straight through: top row 1-20, bottom row 21-44.
TOP ROW, x 50.69 down to 1.27: GND, +3.3V, GND, then CHAN-CLK and CHAN0..CHAN3
(pins 4-8), CHAN4 (pin 9), CHAN5..CHAN11 (10-16), +3.3V, +1.8V, +1.0V, GND.
FOUR positions between pin 9 and pin 10 -- x 26.82, 34.29, 31.75, 29.21 -- are
RESERVED, not spare: the landing for the edge-mounted micro-USB X1, 7.80 mm of
copper in 11.176 mm clear, centred with 1.69 mm each side.
BOTTOM ROW, contiguous, x 50.69 down to 1.27: GND, VU, GND, RST#, CHAN12..CHAN28
(pins 25-41), ANALOG-I00, ANALOG-I01, +5V-INPUT.
FIVE grounds: pins 1, 9, 20, 21, 23. Pin 22-VU SOURCES 5 V, pin 44 only takes it.



Digilent 12-pin Pmod (JA). Pins 1-4 and 7-10 are signals through 200R series resistors; pins 5/11 are GND and 6/12 are VCC3V3 -- so a 2x6 header carries EIGHT signals, not twelve, which is why the names skip JA5/JA6. No decoupling here: the pmod spec puts bypassing on the module, and VCC3V3 already carries ~170uF. The Cmod A7 has none at its Pmod either. JA1-JA4 (the top row, the half a Pmod uses for SPI) are all in bank 16: A14 A15 B15 A16. JA7 is C15, also bank 16. JA8/JA9/JA10 are W16/W27/W7 -- banks 14/35/34 -- because nothing else was free. All five banks are VCC0=3.3V, so this costs timing coherence, not levels.



General IO			
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[illegible][illegible]

Figure 10 shows the pin connections for the AS4C32M1658-7T CN. The diagram is divided into two main sections: a top section for pins DQ0 through DQ15 and a bottom section for pins DQ16 through DQ31. The top section is connected to a 3.3V supply, and the bottom section is connected to a 1.8V supply. The pins are labeled as follows:

- Top Section (3.3V):** DQ0, DQ1, DQ2, DQ3, DQ4, DQ5, DQ6, DQ7, DQ8, DQ9, DQ10, DQ11, DQ12, DQ13, DQ14, DQ15.
- Bottom Section (1.8V):** DQ16, DQ17, DQ18, DQ19, DQ20, DQ21, DQ22, DQ23, DQ24, DQ25, DQ26, DQ27, DQ28, DQ29, DQ30, DQ31.

The diagram also shows the connection of the pins to the 3.3V and 1.8V supplies. The 3.3V supply is connected to the top of the pins, and the 1.8V supply is connected to the bottom of the pins. The diagram is labeled '3.3V' and '1.8V' at the top.

Memory			
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